

HACKATHON PROBLEM STATEMENTS

Hack'oVerse 2025 – Explore the universe of tech innovation.

Web Development

- **College Event Management App/Website (Problem Statement (PS) id: FCAHV01)**
Build a platform to manage college events — allowing event creation, registration, notifications, and result announcements seamlessly.
- **Student Feedback Portal/Website (PS id: FCAHV02)**
Develop a web-based portal where students can securely share feedback on courses, faculty, or events, and administrators can analyse responses.
- **Simple To-Do List Web App (PS id: FCAHV03)**
Create a minimal yet functional task manager that allows users to add, update, and track daily tasks efficiently.
- **Basic Quiz App (PS id: FCAHV04)**
Design a web quiz platform supporting multiple-choice questions, timer functionality, scoring, and leader boards.
- **Online Attendance Management System (PS id: FCAHV05)**
Build a system to record, monitor, and generate attendance reports for students or employees with secure authentication.
- **College Library Management App/Website (PS id: FCAHV06)**
Develop a digital library interface that enables students to browse, issue, and return books while managing inventory online.
- **Mini E-Commerce Website (PS id: FCAHV07)**
Create a basic e-commerce website supporting product listings, shopping cart, order management, and payment simulation.
- **Campus Navigation App (PS id: FCAHV08)**
Design an interactive web-based map that helps new students or visitors navigate campus buildings and facilities.
- **Event & Ticket Booking App (PS id: FCAHV010)**
Build a ticketing platform for college or local events, enabling event discovery, seat booking, and digital ticket generation.

Cybersecurity

- **Password Strength Checker (PS id: FCAHV011)**
Create a tool that evaluates password security using entropy and pattern analysis, and suggests stronger alternatives.
- **Simple Encryption/Decryption Tool (PS id: FCAHV012)**
Build a web or desktop utility that encrypts and decrypts text/files using basic

cryptographic algorithms.

- **Phishing URL Classifier using ML (PS id: FCAHV013)**
Develop a machine learning model that detects and flags phishing URLs based on URL structure, domain reputation, and content cues.

Artificial Intelligence & Machine Learning (AI/ML)

- **Spam Email Classifier (PS id: FCAHV014)**
Build an ML model that distinguishes spam from genuine emails using natural language processing and keyword patterns.
- **Fake News Detection (PS id: FCAHV015)**
Develop a system that classifies news articles as real or fake using NLP and source reliability checks.
- **Sentiment Analysis on Student Feedback (PS id: FCAHV016)**
Analyse student feedback data to classify sentiments (positive, negative, neutral) and visualize overall satisfaction trends.
- **Chatbot for Student Queries (PS id: FCAHV017)**
Build an AI chatbot that answers academic, administrative, or event-related queries for students in real time.
- **AI-Powered Knowledge Hub (PS id: FCAHV018)**
Build a system where users upload notes/documents and can query them conversationally - like a private ChatGPT for personal data.
- **Context-Aware Smart Meeting Assistant (PS id: FCAHV019)**
Create an AI assistant that listens to a meeting transcript and automatically generates a summary, action items, and deadlines for each participant.
- **AI Resume Builder with Skill Gap Analysis (PS id: FCAHV020)**
Analyze resumes against job descriptions and suggest missing skills or online learning recommendations.
- **Smart News Verifier (PS id: FCAHV021)**
Develop an NLP tool that classifies news articles as fake or genuine based on content patterns and source credibility.

Healthcare / MedTech

- **AI-Based Health Risk Predictor (PS id: FCAHV022)**
Predict potential diseases (like diabetes or heart risk) from lifestyle and vital data using a ML model with public datasets.
- **Remote Patient Monitoring Dashboard (PS id: FCAHV023)**
Simulate IoT data (heart rate, oxygen levels, temperature) and build a real-time

dashboard that raises alerts for abnormal readings.

- **Smart Medicine Interaction Checker (PS id: FCAHV024)**
Input prescribed medicines and get alerts about harmful drug interactions using a medical database or public API.
- **Mental Wellness Recommendation Engine (PS id: FCAHV025)**
Use sentiment analysis on daily journal entries to provide personalized mindfulness exercises or stress management tips.

Block chain & Web3

- **Decentralized Freelance Marketplace (PS id: FCAHV026)**
Create a app where freelancers and clients interact, and payments are held in smart contracts until both parties approve delivery.
- **DAO for Community Projects (PS id: FCAHV027)**
Build a DAO system where community members can propose ideas, vote, and allocate funds transparently.
- **Block chain-Based Certificate Verification (PS id: FCAHV028)**
Institutions issue certificates on-chain that can be verified via transaction hash or QR code.
- **Transparent Donation Tracker (PS id: FCAHV029)**
A block chain-based platform where donors can trace the path of their contribution from source to end-user.

Full Stack / Web & App Development

- **Smart Campus Management System (PS id: FCAHV030)**
One portal for college students to track attendance, register for events, access notices, and manage canteen/bus schedules.
- **Collaborative Code Interview Platform (PS id: FCAHV031)**
A live coding web app (like a mini Coder Pad) with chat, problem editor, and automatic code evaluation.
- **Community Crisis Response Platform (PS id: FCAHV032)**
A web app where users report emergencies (like blood needs or local issues), and nearby volunteers get real-time alerts.
- **Task Flow – Team Task Tracker (PS id: FCAHV033)**
A minimal Trello-like task management app for small teams with progress tracking and role-based access.

Artificial Intelligence & Machine Learning (AI/ML) – More to select

- The rise of e-commerce makes it harder for consumers to differentiate between genuine and fake products. [Develop a block chain-based system integrating QR/barcodes for product authenticity, transparency, and real-time verification, empowering consumers and preventing counterfeiting with secure, tamper-proof tracking and decentralized data storage. **(PS id: FCAHV034)**
- Developing an AI-based malaria detection model using prepared blood samples, integrated with a website, to provide cost-effective and accessible testing for rural areas **(PS id: FCAHV035)**
- Farmers face substantial crop losses each year due to unpredictable weather conditions, Lack of knowledge on varying water needs for different crops and inefficient irrigation practices. Even when crops are successfully grown, farmers often struggle to receive fair prices in local markets (mandis), impacting their livelihood. **(PS id: FCAHV036)**
- Food wastage is a global issue “lack of awareness on food wastage” **(PS id: FCAHV037)**
- Predict Crop Yields and market price buy using Ai Tool **(PS id: FCAHV038)**
- Healthcare challenges in Rural India **(PS id: FCAHV039)**
- AI-Based Diagnostic Tools for Remote Healthcare in Rural India : How can AI diagnostic systems be developed for rural clinics to detect diseases like tuberculosis and malaria using mobile devices and telemedicine for timely treatment **(PS id: FCAHV040)**
- A web-based simulation tool replicating real interviews with AI-driven question generation and response analysis **(PS id: FCAHV041)**
- An AI powered phishing E-mail detection. **(PS id: FCAHV042)**
- A web-based receipt generator that suggest meals from leftover ingredients, incorporates meal planning and link with local food donation drives to foster sustainability. **(PS id: FCAHV043)**